

Tutorial - Week 5

TG 11 - TG 12

1. Consider the function $f(x) = \frac{x^3}{3} + 2x^2 - 2$.

- (a) Find the equation of the tangent line at the point $(-3, 7)$.
- (b) Find all the points on the curve $y = f(x)$ where the tangent line is perpendicular to the y -axis.

2. Find the polynomial $f(x) = ax^3 + bx^2 + cx + d$ such that its graph

- (a) is tangent at $x = 0$ to the line $y = \frac{1}{2}x - \frac{3}{2}$, and at $x = 1$ to $y = \frac{15}{2}x - \frac{13}{2}$.
- (b) intersects the x -axis at $x = 2$ with slope 13, and the y -axis at $y = 2$ with slope -11 .
- (c) passes through the origin with slope 1, and has horizontal tangent lines at $x = 1$ and $x = -1$.

3. Prove that $\frac{d}{dx}(\cos x) = -\sin x$.

4. Let f and g two differentiable functions at x with $g(x) \neq 0$. Prove that

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{(g(x))^2}.$$

5. Differentiate the following functions in at least two different ways:

- (a) $f(x) = \frac{2x-5}{x}$.
- (c) $f(x) = x \sqrt[3]{x+2}$.
- (e) $f(x) = \ln \left(\frac{1}{x} \right)$.
- (b) $f(x) = (2x+3)^2$.
- (d) $f(x) = \sqrt{e^x}$.
- (f) $f(x) = \cos^2 x - \sin^2 x$.

6. Consider the differential equation $(x^2 - 9)y' + xy = 0$.

- (a) Show that $y = \frac{c}{\sqrt{x^2-9}}$ satisfies the equation.
- (b) Find c such that $y(5) = 1$.

7. Consider the differential equation $x^3 y''' - 2xy' + 4y = 0$. Show that the following functions satisfy the equation:

- (a) $y_1 = x^2$.
- (b) $y_2 = x^2 \ln x$.
- (c) $y = c_1 x^2 + c_2 x^2 \ln x$.

8. Use the properties of the derivative to compute the derivatives of the following functions.

- (a) $f(x) = e^{nx}$ for $n \in \mathbb{Z}$.
- (b) $f(x) = e^{2x} - \frac{3}{x^{1/3}}$.
- (c) $f(x) = x^2 g(x) + \frac{g(x)}{\sin x} + \frac{\sqrt{x} + \tan x}{xg(x)}$ with $g(x)$ differentiable and $g(x) \neq 0$.

9. Find $\frac{d}{dx}(f(x) + f''(x))$ for $f(x) = c_1 e^{3x} + c_2 \cos x + c_3 \sin x$.

10. Let a be a positive real number.

- (a) Show that $\lim_{x \rightarrow +\infty} \left(1 + \frac{a}{x} \right)^x = e^a$.
- (b) Find $\lim_{x \rightarrow 0} (1 + ax)^{\frac{1}{x}}$.

11. If $F(x) = f(xf(xf(x)))$, where $f(1) = 2$, $f(2) = 3$, $f'(1) = 4$, $f'(2) = 5$ and $f'(3) = 6$, find $F'(1)$.

12. Find the derivatives of the following functions.

(a) $f(x) = \ln(\sin^2 x)$.

(c) $f(x) = e^{e^x + \tan^3 \sqrt{x}}$.

(b) $f(x) = 2^{3^{4^x}}$.

(d) $f(x) = \sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}$.

13. Find the derivatives of the functions defined by the following equations, at the given points.

(a) $y^2 - x^4 = 3$, at $(1, 2)$.

(c) $3xy^2 + 5xy = 8$ at $(1, 1)$.

(b) $y^3 + xy + x^2 = 1$, at $(1, 0)$.

(d) $e^{xy \tan^{-1} x} = y$, at $(0, 1)$.

14. Use logarithmic differentiation to find the derivatives of the following functions.

(a) $y = \frac{e^{2x} x^3 \sin x}{\sqrt[3]{x^2 - 1}}$.

(b) $y = (\cos x)^{3^{x^2}}$.

15. (a) Suppose f is a one-to-one differentiable function and its inverse function f^{-1} is also differentiable. Use implicit differentiation to show that

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

provided that the denominator is not 0.

(b) If $f(4) = 5$ and $f'(4) = \frac{2}{3}$, find $(f^{-1})'(5)$.

Definition of the Hyperbolic Functions

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

$$\tanh x = \frac{\sinh x}{\cosh x}.$$

16. Prove the following identities.

(a) $\cosh(-x) = \cosh(x)$.

(d) $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$.

(b) $\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$.

(e) $\cosh^{-1} = \ln(x + \sqrt{x^2 - 1})$.

(c) $\sinh(-x) = -\sinh(x)$.

(f) $\sinh^{-1} = \ln(x + \sqrt{x^2 + 1})$.

17. Find the horizontal asymptotes of $\tanh x$.

18. Prove that $\frac{d}{dx}(\tanh x) = \text{sech}^2 x$ and $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$